

Abstract

Scientific machine learning (ML) for PDEs aims not only to approximate solutions, but to uncover *low-dimensional structures* in families of solutions. These families correspond to solving the PDEs for different initial or boundary conditions (ICs/BCs), and different physical parameters.

We make the manifold assumption explicit and measurable: learn an embedding via a shared PINN body and quantify its intrinsic dimensionality through PCA of the latent-space covariance matrix.

This enables compression, transfer across initial conditions, and physical interpretability.

Testbed — 1D viscous Burgers' equation:

$$\partial_t u + u \partial_x u = \nu \partial_{xx} u, \quad (x, t, \nu) \in [-5, 5] \times [0, 5] \times [10^{-2}, 1]$$

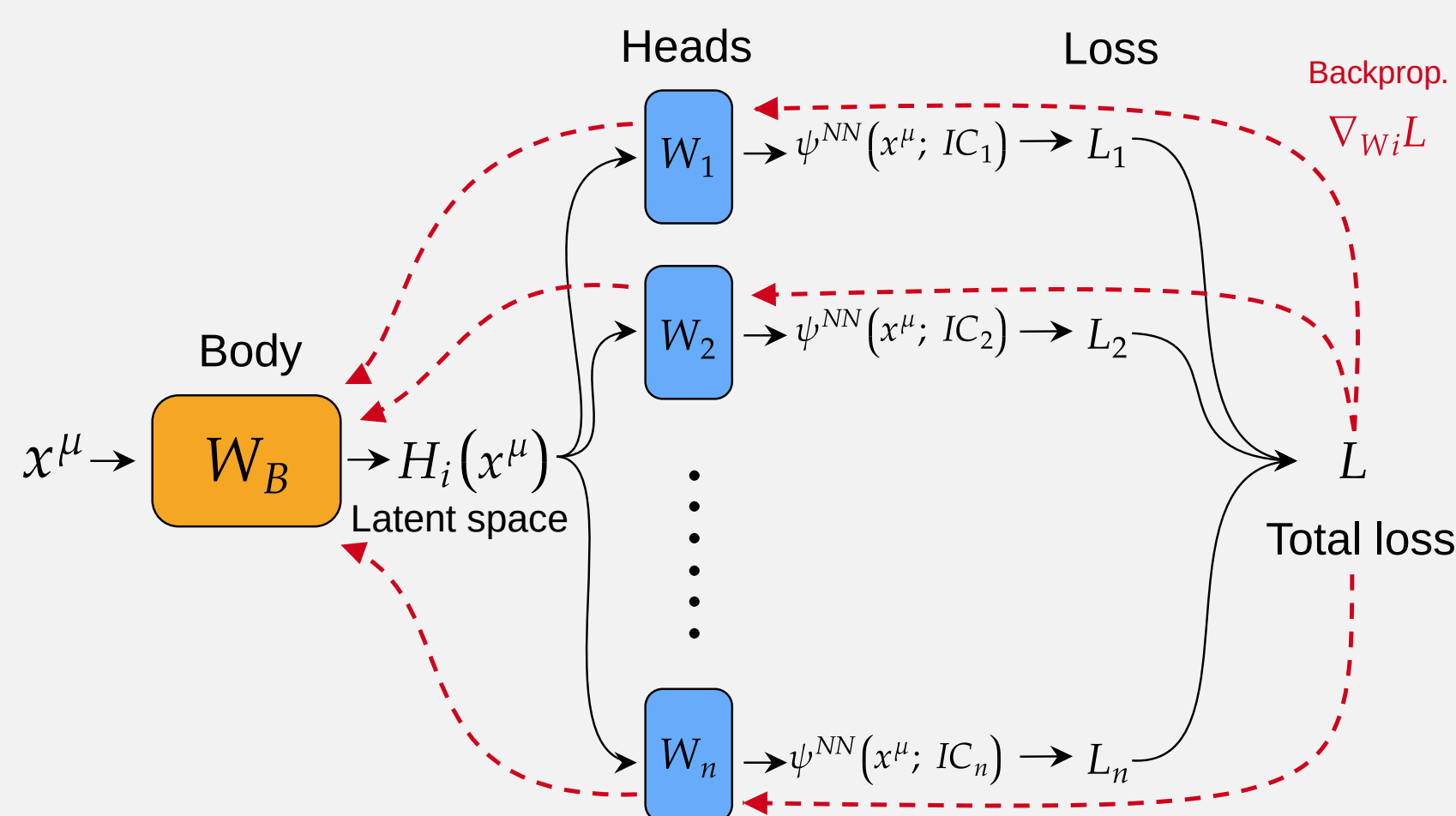
A nonlinear PDE that develops steep gradients and shock-like features as $\nu \rightarrow 0$; a standard surrogate for turbulent transport and multiscale dynamics. We train simultaneously over **25 viscosities** and **20 initial conditions**, and find that the solution space is effectively **3-dimensional**.

Methodology

Instead of learning each solution independently, we learn a **shared functional basis** across initial conditions. We split the network into a **body** producing n_b shared latent functions and **linear heads** that map them to individual solutions:

$$u(x, t, \nu; IC_i) = v_i(x) + (1 - e^{-t}) \sum_{j=1}^{n_b} w_{ij} H_j(x, t, \nu)$$

- $v_i(x)$: enforces the initial condition exactly
- $\{H_j\}$: shared **learned functional basis**
- $\{w_{ij}\}$: coefficients selecting a solution



Training objective (PDE residual + orthogonality penalty):

$$\mathcal{L} = \frac{1}{n_b} \sum_i \frac{1}{M} \sum_m \frac{|\mathcal{R}[u_i(x_m, t_m, \nu_m)]|^2}{\Lambda(\partial_x u)} + \lambda_{\text{ortho}} \mathcal{L}_{\text{ortho}}, \quad \lambda_{\text{ortho}} \sim 10^{-3}$$

Λ is a gradient-weighting factor that stabilizes training near steep gradients as in Ferrer-Sanchez et al. (2024).

The learned embedding is not unique: different runs produce rotated versions of the same subspace. We add a lightweight penalty:

$$\mathcal{L}_{\text{ortho}} = \|WW^T - I\|_F^2 + \|W^T W - I\|_F^2$$

We enforce approximate orthogonality to make the representation stable and comparable.

The explained-variance spectrum provides a robust fingerprint of the PDE solution manifold.

Differences between IC ensembles reflect intrinsic PDE properties, not arbitrary latent rotations.

Results

Training setup (shared across experiments)

Parameter	Value
Body architecture	FCNN, 5 layers $\times$ 128 neurons, tanh
Latent dim. n_b	20
Heads	20 linear layers (one per IC, no bias)
Viscosity range	$\nu \in [10^{-2}, 1]$, 25 log-spaced values
Spatial/time grid	$x \in [-5, 5]$, $t \in [0, 5]$, 100 pts each
Optimiser	Adam $\eta_0 = 10^{-3}$, 1k warm-up, $\times 0.985/1k$
Ortho penalty	$\lambda_{\text{ortho}} = 10^{-3}$ (2 orders below PDE loss)
IC families	Fourier modes & Polynomials (two separate runs)
Hardware	NVIDIA H100 GPU (≈ 5 days/run)

Fourier ICs:

Random Fourier modes with homogeneous BCs

$$v_i(x) = \begin{cases} A_i \sin\left(\frac{n_i \pi x}{5}\right) \\ B_i \cos\left(\frac{m_i \pi x}{10}\right) \end{cases}$$

$$A_i, B_i \sim \mathcal{U}[0.1, 2]$$

Polynomial ICs:

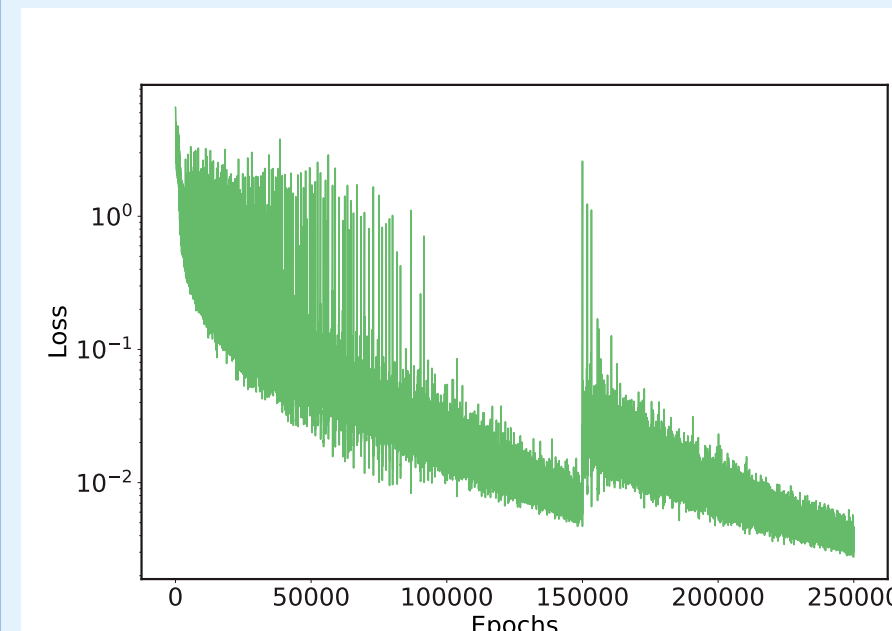
Random low-degree polynomials vanishing at the boundaries

$$v_i(x) = \sum_j C_j^i p^{a_j}(x)$$

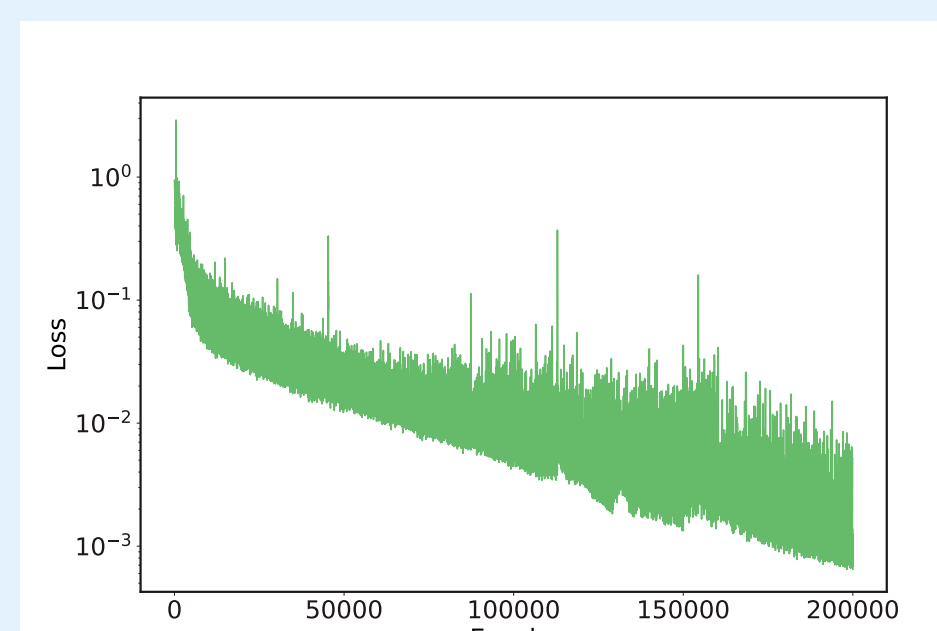
$$a_j \in \{2, \dots, 7\}, C_j^i \sim \mathcal{U}[0, 1]$$

Both IC families exhibit strong dimensional collapse.

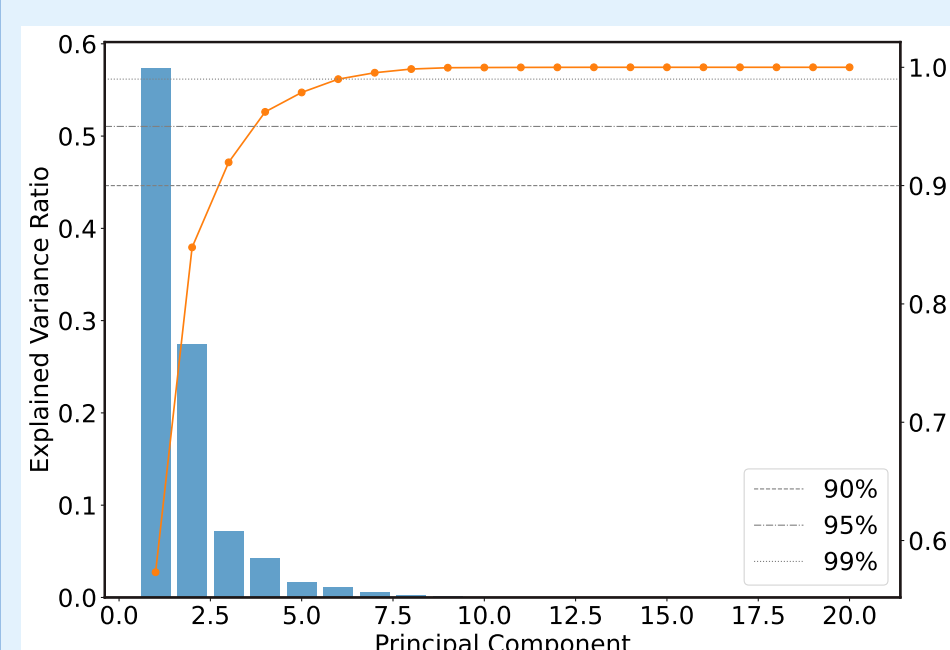
PDE loss — Fourier ICs



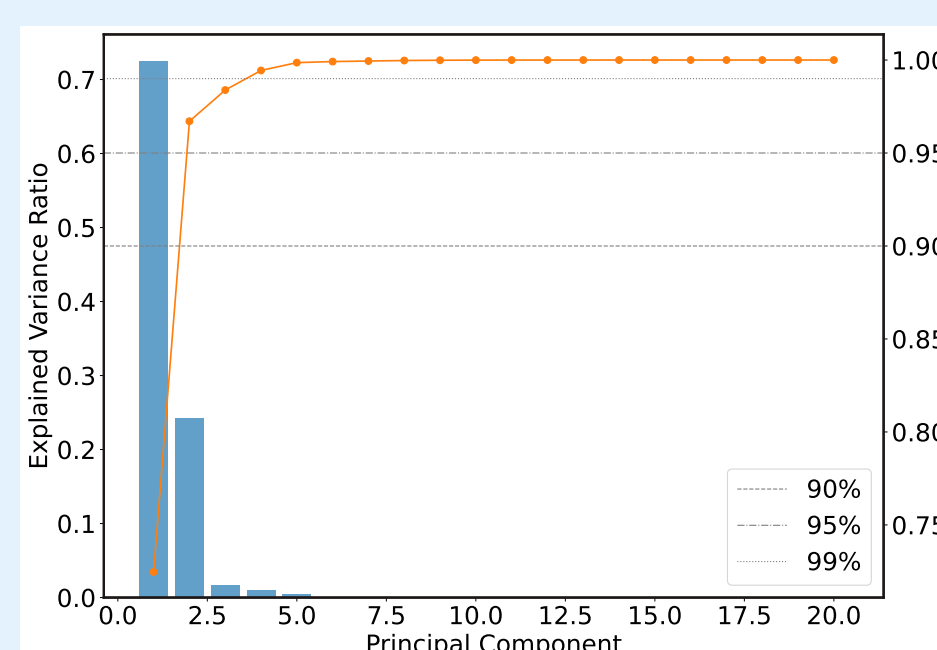
PDE loss — Polynomial ICs



Expl. var. — Fourier ICs



Expl. var. — Polynomial ICs



Bars: individual eigenvalues (left axis). Red curve: cumulative explained variance (right axis).

The solution manifold is effectively 3-dimensional.
Polynomial ICs compress faster, indicating smoother dynamics.

Conclusions

A nonlinear PDE with shocks admits a **low-dimensional latent structure**.

Multiscale organisation:

- Leading components: global, smooth structure (IR)
- Subleading components: localized gradients (UV)
- Truncation provides a simple and effective model-reduction mechanism.

Future directions:

- Define a geometric metric on the latent manifold
- Transfer to unseen regimes via head fine-tuning
- Extend to Navier–Stokes and compare with classical decompositions

Key refs: Raissi et al. (2019); Lu et al. (2021); Ferrer-Sanchez et al. (2024); Tarancón-Álvarez et al. (2025)

